Title Here

Your Name[1]

Abstract

Brief abstract here.

Introduction

Introduction for simulation study.

Motivating Problem

Brief description of the motivation for your simulation study. This could also be a theorem definition/description section depending on what you choose.

Simulation Set-up

Describe your simulation study.

An example of a figure is below. In the Rmd file, the `chunk-name` can be changed to whatever you want so that it can be referenced using `\ref{tab:chunk-name}` (where "tab" represents a table, "fig" is for a figure).

NOTE: the code chunks have `echo=F` to suppress the code from appearing in your limited 2 page report. It can still automatically appear at the end in the appendix with a nifty line of code that appears at the end!

The code chunk(s) to implement your simulation study might be good to include at the end of this section:

Results

Write your results in this section. The interpretation or implications of results will primarily appear in the following Discussion/Conclusion section.

One example you can use to create a table in our PDF output example here is with the `kableExtra` package:

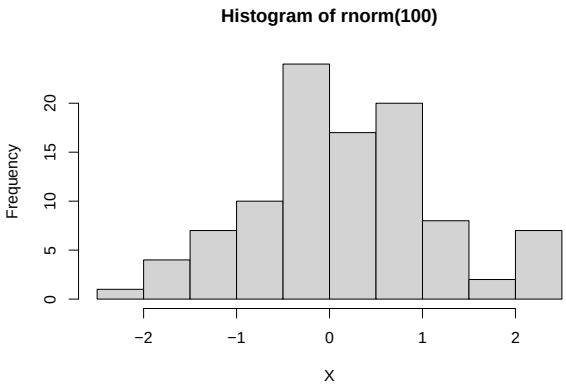


Figure 1: Caption for Figure

Table 1: Caption for table

	Column 1 of Results	Column 2 of Results
Row 1	1	5
Row 2	2	6
Row 3	3	7
Row 4	4	8

Conclusion

Write your conclusion here where you interpret the results for your given context, including any limitations.

[1] Your department/affiliation

References

Code Appendix

```
knitr::opts_chunk$set(echo = TRUE)

library(kableExtra) # load for tables
# Create figure to demonstrate something graphically

set.seed(515) # set seed to reproduce example figure
hist(rnorm(100), xlab='X')
# Implement your simulation(s) in this code chunk

mysim <- function(x){
  # function here for example
}

# Run simulation
#sapply(1:1000, function(x) mysim(x))
# Extract results for null case and scenario with expected 80% power
tab_res <- matrix(1:8, nrow=4, ncol=2) # made up matrix with results
rownames(tab_res) <- c('Row 1','Row 2','Row 3','Row 4')

# Create table of results
kbl(tab_res,
    caption = 'Caption for table',
    booktabs=T, align='cc',
    col.names = c('Column 1 of Results','Column 2 of Results'))  %>%
    collapse_rows(columns=1, latex_hline='major', valign='middle') %>%
    kable_styling(latex_options = "HOLD_position")
```